



Adaptive Interview Intelligence Platform Using Multi-Agent AI and Retrieval-Augmented Generation.

Major Project Review - 1

Batch ID: AIML-A10

Project Guide

Mrs.V. Asha Jyothi
Asssistant Professor,
AIML, VVIT

Project Batch Members

E. ARUN KUMAR – 24BQ5A6103	K. NAGA SUDHA LAKSHMI – 23BQ1A6150
CH. TEJASWINI – 23BQ1A6122	B. HARSHITHA – 23BQ1A6105

Problem Statement

-  Interview preparation is becoming increasingly important for students and job seekers.
-  Candidates need technical knowledge, communication skills and problem-solving ability.
-  Existing preparation methods are often generic and do not adapt to individual candidates.
-  Static question banks provide similar questions to different users.
-  Conventional mock interview platforms provide limited personalization.
-  Candidates may not receive meaningful feedback about their strengths and weaknesses.
-  There is a need for an intelligent system that understands the candidate's resume, target role and previous performance.
-  The system should dynamically adjust questions, difficulty and recommendations.



Core Problem:

"Existing interview preparation systems are mostly generic and do not continuously adapt to the candidate's skills, performance, weaknesses and target job role."

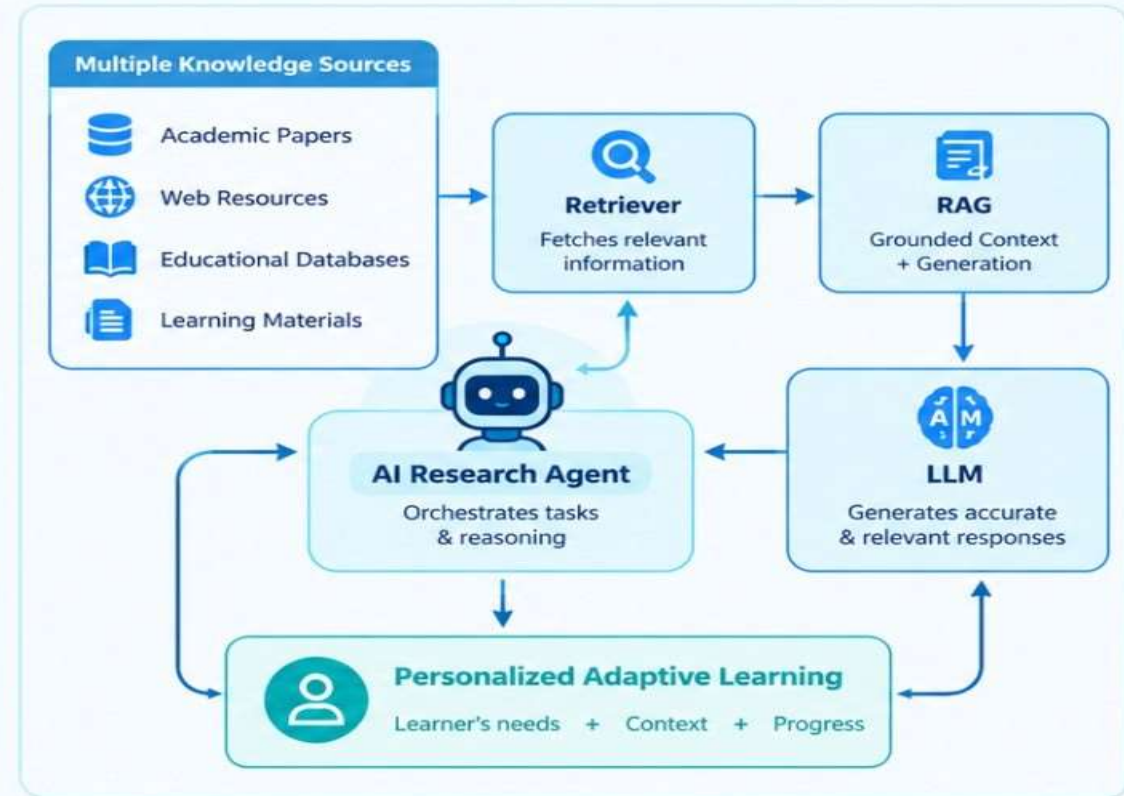
LITERATURE REVIEW

Ref 01 – EduMSRA: A Multi-Source Educational Research Agent Integrating RAG and MCP for Adaptive Intelligent Tutoring Systems

 Author et al., 2026 | MDPI

PROPOSED METHOD

- Multi-source AI research agent
- Retrieval-Augmented Generation (RAG)
- Agent-based educational tasks
- Retrieval from relevant knowledge sources
- Adaptive and personalized learning
- Grounded and relevant AI-generated responses



RAG + AI AGENT + ADAPTIVE LEARNING

LITERATURE REVIEW



Ref 01 – EduMSRA: A Multi-Source Educational Research Agent Integrating Retrieval-Augmented Generation and Model Context Protocol for Adaptive Intelligent Tutoring Systems

EduMSRA | Recent Research | Adaptive Intelligent Tutoring Systems



DATA & SETUP



Application Domain:

Education and adaptive intelligent tutoring



Target Users:

Learners/students using personalized educational support



Data Sources:

Multiple educational and knowledge sources



Knowledge Retrieval:

Relevant information is retrieved from external knowledge sources before response generation



AI Setup:

AI agent integrated with Retrieval-Augmented Generation (RAG)



Adaptive Setup:

Learner needs and context are considered to provide personalized learning support



Architecture:

Multi-source retrieval → AI agent → information processing → personalized response



Important Note:

The supplied abstract does not specify a particular dataset name, dataset size, or train/test split, so these details are not added.



Educational / Knowledge
Sources



Information Retrieval



RAG + AI Agent



Learner Context



Personalized Learning
Response

LITERATURE REVIEW



Ref 01 – EduMSRA: A Multi-Source Educational Research Agent Integrating Retrieval-Augmented Generation and Model Context Protocol for Adaptive Intelligent Tutoring Systems

EduMSRA | Recent Research | Adaptive Intelligent Tutoring Systems

★ KEY RESULTS



Personalization

The system provides learning support based on the learner's needs and context.



Relevance

RAG retrieves relevant information from multiple knowledge sources before generating responses.



Reliability

Using retrieved knowledge helps the AI provide more accurate and context-relevant answers.



Adaptive Learning

The system can adjust its educational assistance according to the learner's requirements.



Multiple Knowledge Sources



RAG Retrieval



AI Agent Processing



Learner Context



Personalized & Relevant Response



KEY TAKEAWAY

“The study shows that combining RAG, multiple knowledge sources and AI-agent-based processing can provide more relevant, reliable and personalized learning support.”

LITERATURE REVIEW



Ref 01 – EduMSRA: A Multi-Source Educational Research Agent Integrating Retrieval-Augmented Generation and Model Context Protocol for Adaptive Intelligent Tutoring Systems

EduMSRA | Recent Research | Adaptive Intelligent Tutoring Systems



STRENGTHS



Multi-Source Knowledge

Uses information from multiple educational and knowledge sources.



RAG-Based Retrieval

Retrieves relevant information before generating responses.



Personalized Learning

Adapts responses according to learner needs and context.



AI-Agent Approach

Uses an intelligent agent to process retrieved information and support learning.



Better Reliability

Retrieval of relevant information helps provide more accurate and useful responses.



Adaptive Support

Can support personalized and context-aware learning.



LIMITATIONS



Educational Focus

Mainly designed for adaptive tutoring and educational applications.



Dependence on Knowledge Sources

Quality of responses depends on the availability and quality of retrieved information.



Limited Interview Focus

Does not specifically address technical/HR interview preparation.



Limited Candidate Evaluation

Does not focus on detailed interview answer evaluation and performance analytics.



Domain-Specific Adaptation

The approach is primarily centered on learners rather than job roles and recruitment needs.



Human Decision Support

It supports learning but is not designed to replace human decision-making in recruitment.



KEY TAKEAWAY

“EduMSRA provides a strong foundation for adaptive and personalized intelligent tutoring using multi-source knowledge, RAG and AI agents, but is limited to educational use cases and does not address technical/HR interview preparation or detailed candidate evaluation.”

LITERATURE REVIEW



Ref 01 – EduMSRA: A Multi-Source Educational Research Agent Integrating Retrieval-Augmented Generation and Model Context Protocol for Adaptive Intelligent Tutoring Systems



RELEVANCE TO MY RESEARCH



RAG Integration

Provides the idea of retrieving relevant information before generating AI responses.



Multi-Agent Approach

Supports our use of multiple AI agents for different interview preparation tasks.



Personalization

The adaptive learning concept can be applied to personalize interview questions and recommendations.



Context-Aware Responses

Retrieved knowledge and candidate context can help generate more relevant interview interactions.



Reliable AI Feedback

RAG-based retrieval can help improve the reliability and relevance of AI-generated feedback.



Adaptive Interview Preparation

The learning adaptation concept can be extended to adjust interview difficulty based on the candidate's performance.



KEY CONNECTION

EduMSRA



Adaptive Learning +
RAG + AI Agents



Our Research



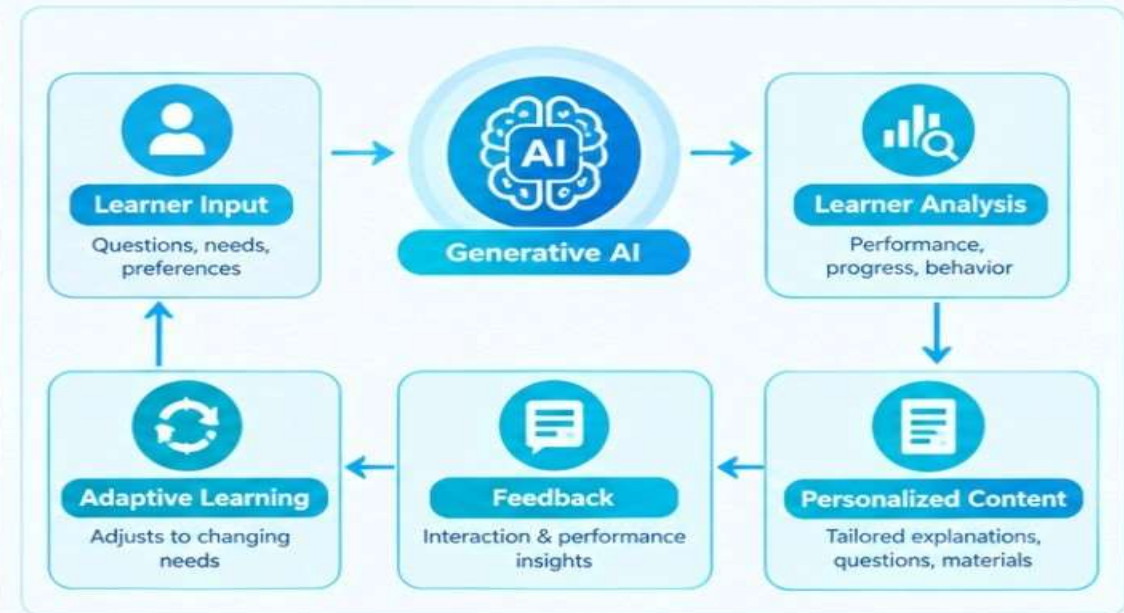
Adaptive Interview
Preparation + RAG +
Multi-Agent AI +
Personalized Feedback

LITERATURE REVIEW

Ref 02 – Bringing Generative AI to Adaptive Learning in Education

Ref 02 | Recent Research | Adaptive Learning | Generative AI in Education

PROPOSED METHOD



METHOD FLOW



LITERATURE REVIEW

Ref 02 – Bringing Generative AI to Adaptive Learning in Education

Ref 02 | Recent Research | Adaptive Learning | Generative AI in Education

DATA & SETUP

-  **Target Data**
Learner information, learning progress and performance.
-  **Learner Interaction**
Uses learner inputs and interactions with the AI system.
-  **Learning Content**
Educational content and learning materials are used for generating personalized learning experiences.
-  **AI Setup**
Generative AI is used to create and adapt learning content dynamically.
-  **Adaptive Setup**
The system considers individual learner needs and performance to modify the learning experience.
-  **Feedback Data**
Learner performance and feedback help guide future learning activities.

SETUP FLOW



LITERATURE REVIEW

Ref 02 – Bringing Generative AI to Adaptive Learning in Education Bringing Generative AI to Adaptive Learning in Education

KEY RESULTS

-  **Personalized Learning**
Generative AI can provide learning content based on individual learner needs. ✓
-  **Adaptive Content**
Learning materials can be changed according to the learner's progress and performance. ✓
-  **Better Learner Engagement**
AI-based interactive learning can make the learning process more engaging. ✓
-  **Continuous Adaptation**
The system can continuously adjust the learning experience based on learner feedback and interactions. ✓

-  **Improved Feedback**
Learner performance can be used to provide more relevant feedback and recommendations. ✓
-  **Dynamic Content Generation**
Generative AI can create customized questions, explanations, and learning materials instead of using only fixed content. ✓
-  **Individual Learning Support**
Different learners can receive different content according to their abilities and requirements. ✓

KEY TAKEAWAY

“
Generative AI combined with adaptive learning can create a more **personalized, flexible, and interactive** learning experience.
”

KEY CONCEPT FLOW



LITERATURE REVIEW

Ref 02 – Bringing Generative AI to Adaptive Learning in Education Bringing Generative AI to Adaptive Learning in Education

STRENGTHS

-  **Personalized Learning**
Adapts learning to individual learner needs. 
-  **Dynamic Content**
Generates customized questions, explanations, and materials. 
-  **Adaptive Learning**
Adjusts content based on learner progress and performance. 
-  **Interactive Learning**
Supports more engaging AI-based learner interactions. 
-  **Continuous Feedback**
Uses learner performance to improve future activities. 
-  **Flexible Learning**
Supports different learning requirements and abilities. 

LIMITATIONS

-  **Education Focus**
Mainly designed for educational learning environments. 
-  **Domain Dependency**
Adaptation depends on the available learning content. 
-  **Limited Interview Use**
Does not directly address interview preparation. 
-  **Limited Evaluation**
Focuses on learning adaptation rather than candidate answer evaluation. 
-  **No Recruitment Context**
Does not support actual recruitment or hiring decisions. 
-  **Human Oversight**
AI-generated learning still requires appropriate human guidance. 



KEY TAKEAWAY

“Generative AI combined with adaptive learning can create a more **personalized, flexible, and interactive learning experience.**”

LITERATURE REVIEW



Ref 02 – Relevance to My Research

Bringing Generative AI to Adaptive Learning in Education



Adaptive Learning

The paper shows how AI can adapt learning based on individual performance and needs. Our system applies the same idea to adapt interview questions according to candidate performance.

01



Personalized Experience

Generative AI can provide personalized content. Our platform personalizes interview preparation for each candidate based on their resume, skills, weaknesses, and target role.

02



Dynamic Question Generation

The paper supports dynamically generated learning content. Our research extends this concept to generate customized technical and HR interview questions.

03



Performance-Based Adaptation

Learner performance is used to decide the next learning activity. Similarly, our platform uses interview performance to adjust question difficulty and future recommendations.

04



Continuous Feedback

The paper highlights feedback-driven learning. Our system provides continuous feedback on candidate answers, communication, and technical performance.

05



KEY CONNECTION



Adaptive Learning

+



Generative AI

→



Adaptive Interview Preparation
+ Personalized Feedback

LITERATURE REVIEW



Ref 03 – A Survey on Retrieval-Augmented Text Generation for Large Language Models

Ref 03 | Recent Research | Retrieval-Augmented Generation (RAG)



PROPOSED METHOD



Retrieval-Augmented Generation (RAG)

Retrieves relevant information before generating an answer.



External Knowledge Retrieval

Uses external knowledge sources to provide additional information to the language model.



Query-Based Retrieval

The user's query is used to find the most relevant information.



Retrieved Context

The relevant retrieved information is added as context for the language model.



Context-Aware Generation

The model generates answers using both the user query and retrieved information.



Improved Reliability

Retrieval helps provide more relevant and knowledge-grounded responses.



METHOD FLOW



User Query



Information Retrieval



Relevant Context



LLM



Generated Response

LITERATURE REVIEW



Ref 03 – A Survey on Retrieval-Augmented Text Generation for Large Language Models

Ref 03 | Recent Research | Retrieval-Augmented Generation (RAG)



DATA & SETUP



Input

User query or question given to the language model.



Knowledge Sources

External documents or knowledge sources are used for retrieval.



Retrieval Data

Relevant information is searched from the available knowledge base.



Retrieved Context

The most useful information is selected and given to the language model.



AI Model

Large Language Model (LLM) is used to generate the final response.



RAG SETUP



User Query

(question / input)



Retriever

(searches knowledge sources)



Relevant Documents

(filtered & relevant info)



LLM

(generates response)



Final Answer

(knowledge-grounded response)



MAIN PURPOSE

Combine external knowledge with LLM generation to produce **more relevant** and **reliable** responses.

LITERATURE REVIEW



Ref 03 – A Survey on Retrieval-Augmented Text Generation for Large Language Models

Key Results



KEY RESULTS



Better Information Retrieval

RAG helps find relevant information from external knowledge sources.



More Relevant Answers

Retrieved information helps the LLM generate answers related to the user's query.



Knowledge-Grounded Responses

The generated response is supported by retrieved information rather than only the model's internal knowledge.



Reduced Hallucination

Using external information can reduce incorrect or unsupported responses.



Improved LLM Performance

Combining retrieval with generation improves the usefulness of language-model responses.



Updated Knowledge

External sources allow the system to use information that may not be available in the original model knowledge.



Flexible Application

RAG can be applied to different domains where reliable external knowledge is required.



KEY TAKEAWAY



RAG

+



LLM

→



More Relevant, Knowledge-Grounded and Reliable Responses

LITERATURE REVIEW



Ref 03 – A Survey on Retrieval-Augmented Text Generation for Large Language Models

Strengths & Limitations



STRENGTHS



Relevant Knowledge

Retrieves useful information for the given query.



Better Accuracy

External knowledge can improve the quality of generated answers.



Reduced Hallucination

Retrieved context helps reduce unsupported responses.



Updated Information

Can use information from external knowledge sources.



Flexible Approach

RAG can be used for different applications and domains.



Better Context

Gives the LLM additional information before generating the answer.



LIMITATIONS



Retrieval Quality

Poor retrieval can lead to poor final answers.



Knowledge Source Dependency

Performance depends on the quality of external data.



Extra Processing

Retrieval adds an additional step before answer generation.



Higher Complexity

RAG systems are more complex than using an LLM alone.



Domain Adaptation

Different domains may require different knowledge sources.



Not Fully Error-Free

Retrieved information can still be incorrect or incomplete.



KEY TAKEAWAY



RAG + LLM



More Relevant, Knowledge-Grounded and Reliable Responses

LITERATURE REVIEW



Ref 03 – Relevance to My Research

Connecting RAG Concepts with Our Adaptive Interview Preparation System



RAG Integration

Our platform uses RAG to retrieve relevant information before generating interview responses.



Reliable Feedback

Retrieved knowledge can help provide more relevant and reliable feedback to candidates.



Knowledge-Based Interview

External knowledge sources can support technical interview questions and answers.



Context-Aware Responses

Retrieved information can be combined with candidate details to generate context-aware interview interactions.



Reduced Hallucination

RAG can reduce unsupported AI responses, improving the reliability of interview evaluation.



Personalized Preparation

Relevant retrieved information can help generate questions and recommendations based on the candidate's requirements.



KEY CONNECTION



Ref 03

RAG + External Knowledge + Reliable Generation



Our Research

RAG + Multi-Agent AI + Adaptive Interview Preparation

Literature Review - Summary

Reference	Year	Main Idea	Strengths	Limitations
EduMSRA	2026	AI research agent using RAG and multiple knowledge sources for adaptive learning	Personalized, RAG-based, reliable responses	Mainly focused on education; limited interview application
Bringing Generative AI to Adaptive Learning in Education	2024	Uses Generative AI and adaptive learning to personalize learning content	Dynamic content, personalization, continuous feedback	Education-focused; limited recruitment/interview context
A Survey on Retrieval-Augmented Text Generation for LLMs	2024	Studies RAG techniques for improving LLM-generated responses	Relevant knowledge, better context, reduced hallucination	Depends on retrieval quality and external knowledge
Retrieval-Augmented Generation for AI-Generated Content: A Survey	2024	Reviews RAG approaches for knowledge-grounded AI generation	Better factuality, external knowledge, flexible applications	Retrieval complexity; quality depends on knowledge sources

THANK YOU



Questions?



Adaptive Interview Intelligence Platform
Using Multi-Agent AI and Retrieval-Augmented Generation



Major Project Review – 1



Batch ID: AIML-A10